2024 AMC 10A Problem 20 - Solution

Review the full statement and step-by-step solution for 2024 AMC 10A Problem 20. Great practice for AMC 10, AMC 12, AIME, and other math contests

2024 AMC 10A Problem 20

Problem:

Let S be a subset of $\{1, 2, 3, \ldots, 2024\}$ such that the following two conditions hold:

- If x and y are distinct elements of S, then $|x-y|>2$.
- If x and y are distinct odd elements of S, then $|x-y|>6$.

What is the maximum possible number of elements in S ?

Answer Choices:

A. 436

B. 506

C. 608

D. 654

E. 675

Solution:

If S consists of the positive integers less than or equal to 2024 that are congruent to 1,4 , or 8 modulo 10 , then every pair of elements in S differ by at least $|4-1|=|11-8|=3$, and every pair of odd elements of S differ by at least $|11-1|=10$. This set,

$$\{1,4,8,11,14,18,\ldots,2011,2014,2018,2021,2024\}$$

satisfies the given conditions and has $3\cdot\left(\frac{2020}{10}\right)+2=\text{(C)}\boxed{608}$ elements. To see that no larger set satisfies the given conditions, note that if a set satisfies the first condition and some block of 10 consecutive integers contains 4 elements of the set, then those 4 elements would need to be the $1\text{st}, 4$ th, 7 th, and 10th elements in that block, and the two odd numbers among them would be differ by 6 , in violation of the second condition. Therefore there are at most $3\cdot 202=606$ elements of S among the first 2020 positive integers, and at most 2 elements of S can be among $\{2021,2022,2023,2024\}$.